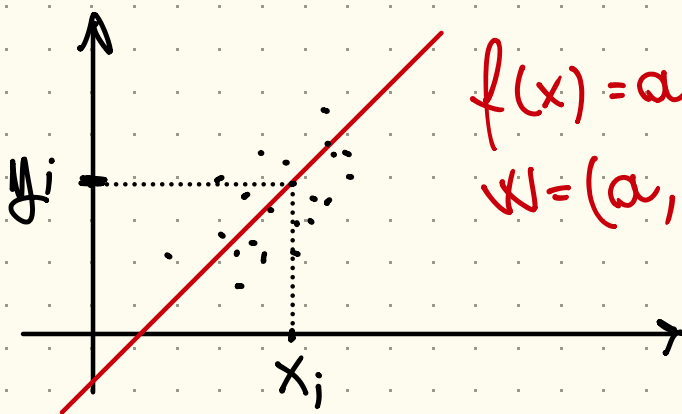


MASSIMI E MINIMI(AI)

$(x_i, y_i)_{i=1}^N$ TRAINING SET

- CERCARE UNA RETE NEURALE f_W CON $W \in \mathbb{R}^d$ PRESI T.C. $f_W(x_i) \approx y_i$

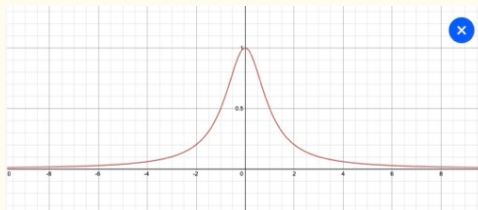
$$\min \sum_{i=1}^N |f_W(x_i) - y_i|^2$$



$$f(x) = ax + b$$
$$W = (a, b) \in \mathbb{R}^2$$

$$\min_{a, b \in \mathbb{R}} \sum_{i=1}^N |f_W(x_i) - y_i|^2$$

es) $f(x) = \frac{1}{x^2 + 1}$



$$\bullet \max_{x \in \mathbb{R}} \frac{1}{x^2 + 1} = 1$$

• $x = 0$ È UN PUNTO DI MAX PER f

$$\cdot \exists \min_{x \in \mathbb{R}} f(x)$$

$$\cdot \inf_{x \in \mathbb{R}} f(x) = 0$$

DERIVATE

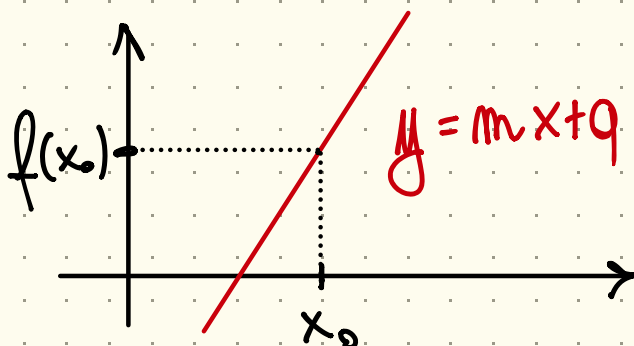
• IL COEFFICIENTE ANGOLARE DELLA SECANTE È:

$$\frac{f(x_0 + h) - f(x_0)}{x_0 + h - x_0} = \frac{f(x_0 + h) - f(x_0)}{h}$$

↑
RAPPORTO INCREMENTALE

• IL COEFFICIENTE ANGOLARE DELLA TANGENTE È:

$$m = \lim_{h \rightarrow 0} m = \frac{f(x_0 + h) - f(x_0)}{h}$$



DEF $f: I \rightarrow \mathbb{R}, x_0 \in I$
INTERVALLO

f SI DICE DERIVABILE IN x_0 SE

$$\exists \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} \in \mathbb{R}$$

$f'(x_0) \leftarrow$ DERIVATA DI f IN x_0

b) $f(x) = x^2$

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0+h) - f(x_0)}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{(x_0+h)^2 - x_0^2}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{x_0^2 + 2hx_0 + h^2 - x_0^2}{h} =$$

$$= \lim_{h \rightarrow 0} \frac{\cancel{h}(2x_0+h)}{\cancel{h}} =$$

$$= \lim_{h \rightarrow 0} 2x_0 + h = 2x_0$$

DEF. $f: I \rightarrow \mathbb{R}$ SI DICE DERIVABILE SE:

- È DERIVABILE IN $x_0 \forall x_0 \in I$

• LA FUNZIONE:

- $f': I \rightarrow \mathbb{R}, x \rightarrow f'(x)$

• SI CHIAMA DERIVATA PRIMA:

- $f(x) = x^2 \Rightarrow f'(x) = 2x$

es) $f: [0, +\infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x}$

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} =$$
$$= \lim_{x \rightarrow x_0} \frac{\overset{(a-b)}{\sqrt{x} - \sqrt{x_0}} \cdot \overset{(a+b)}{\sqrt{x} + \sqrt{x_0}}}{x - x_0} =$$

$$= \lim_{x \rightarrow x_0} \frac{(\sqrt{x})^2 - (\sqrt{x_0})^2}{(x - x_0)(\sqrt{x} + \sqrt{x_0})} =$$

$$= \lim_{x \rightarrow x_0} \frac{x - x_0}{(x - x_0)(\sqrt{x} + \sqrt{x_0})} =$$

$$= \lim_{x \rightarrow x_0} \frac{1}{\sqrt{x} + \sqrt{x_0}} = \begin{cases} \frac{1}{2\sqrt{x_0}} & x_0 > 0 \\ +\infty & x_0 = 0 \end{cases}$$

